Android Programming

Practical programs

1. Hello World Program (Write a program to Toast Hello World)

Xml File

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >

<Button
    android:id="@+id/button1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_centerHorizontal="true"
    android:layout_marginLeft="52dp"
    android:layout_marginTop="61dp"
    android:text="SHOW TOAST" />

</RelativeLayout>
```

Java File

```
package com.andoubleos.myprogram;

import android.os.Bundle;
import android.app.Activity;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.Toast;


public class MainActivity extends Activity {
      Button btn;
      @Override
      protected void onCreate(Bundle savedInstanceState) {
            super.onCreate(savedInstanceState);
            setContentView(R.layout.activity_main);

        btn=(Button) findViewById(R.id.button1);

        btn.setOnClickListener(new OnClickListener() {

                  @Override
                  public void onClick(View arg0) {
                        // TODO Auto-generated method stub
                        Toast.makeText(MainActivity.this, "Hello World",
Toast.LENGTH_LONG).show();
                  }
            });
      }}
```

OUTPUT

2. Addition of two Numbers (Write a program to add two numbers)

Xml File

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >

<TextView
      android:id="@+id/textView1"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_alignParentLeft="true"
      android:layout_below="@+id/textView1"
      android:layout_marginLeft="20dp"
      android:layout_marginTop="43dp"
      android:text="Enter First No"/>
```

```
<EditText
      android:id="@+id/editText1"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_alignBottom="@+id/textView2"
      android:layout_alignRight="@+id/textView1"
      android:ems="2"
      android:inputType="numberDecimal" >

<TextView
      android:id="@+id/textView2"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_alignParentLeft="true"
      android:layout_below="@+id/textView1"
      android:layout_marginLeft="20dp"
      android:layout_marginTop="43dp"
      android:text =" Enter Second No "/>
<EditText
      android:id="@+id/editText2"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_alignBottom="@+id/textView2"
      android:layout_alignRight="@+id/textView1"
      android:ems="2"
      android:inputType="numberDecimal" >
 <TextView
     android:id="@+id/textView3"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_alignLeft="@+id/textView2"
      android:layout_below="@+id/textView2"
      android:layout_marginTop="47dp"
      android:text=" " />
 <Button
       android:id="@+id/button1"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_alignParentBottom="true"
       android:layout_centerHorizontal="true"
       android:layout_marginBottom="22dp"
       android:text="Show SUM" />

</RelativeLayout>
```

Java File

```
package com.example.twosum;

import android.os.Bundle;
import android.app.Activity;
import android.view.Menu;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

     public class MainActivity extends Activity {

      EditText firstNumber;
      EditText secondNumber;
      TextView addResult;
      Button btnAdd;
```

```
  double num1,num2,sum;
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    firstNumber = (EditText)findViewById(R.id. editText1);
    secondNumber = (EditText)findViewById(R.id. editText2);
    addResult = (TextView)findViewById(R.id. textView3);
    btnAdd = (Button)findViewById(R.id.button1);

    btnAdd.setOnClickListener(new OnClickListener() {

    public void onClick(View v) {
       num1 = Double.parseDouble(firstNumber.getText().toString());

       num2 = Double.parseDouble(secondNumber.getText().toString());
       sum = num1 + num2;
       addResult.setText(Double.toString(sum));

         }
     });
  } }
```

OUTPUT

3. Fetch data from an EditText and display it in a TextView

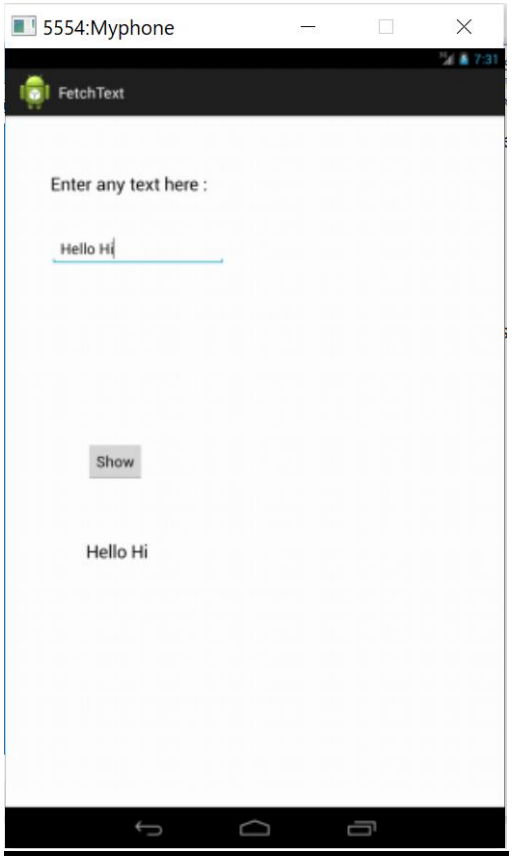
Xml File

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >

    <EditText
        android:id="@+id/editText1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignLeft="@+id/textView1"
        android:layout_below="@+id/textView1"
        android:layout_marginTop="44dp"
        android:ems="10" />

    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentTop="true"
        android:layout_marginLeft="39dp"
        android:layout_marginTop="51dp"
        android:text="Enter any text here :"
        android:textAppearance="?android:attr/textAppearanceLarge" />

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignLeft="@+id/editText1"
        android:layout_centerVertical="true"
        android:layout_marginLeft="43dp"
        android:text="Show" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignLeft="@+id/button1"
        android:layout_below="@+id/button1"
        android:layout_marginTop="69dp"
        android:text=" "
        android:textAppearance="?android:attr/textAppearanceLarge" />

    </RelativeLayout>
```

Java File

```
package com.example.fetchtext;
import android.os.Bundle;
import android.app.Activity;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends Activity {

      EditText e1;
      TextView t1;
      Button b1;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        e1=(EditText) findViewById(R.id.editText1);
        t1=(TextView) findViewById(R.id.textView2);
        b1=(Button) findViewById(R.id.button1);
        b1.setOnClickListener(new OnClickListener() {

                @Override
                public void onClick(View arg0) {
                      // TODO Auto-generated method stub
                      t1.setText(e1.getText().toString());
                                  }
            });
    }}
```

OUTPUT



4. Multiplication Table (Write a program to display multiplication table of a given number)

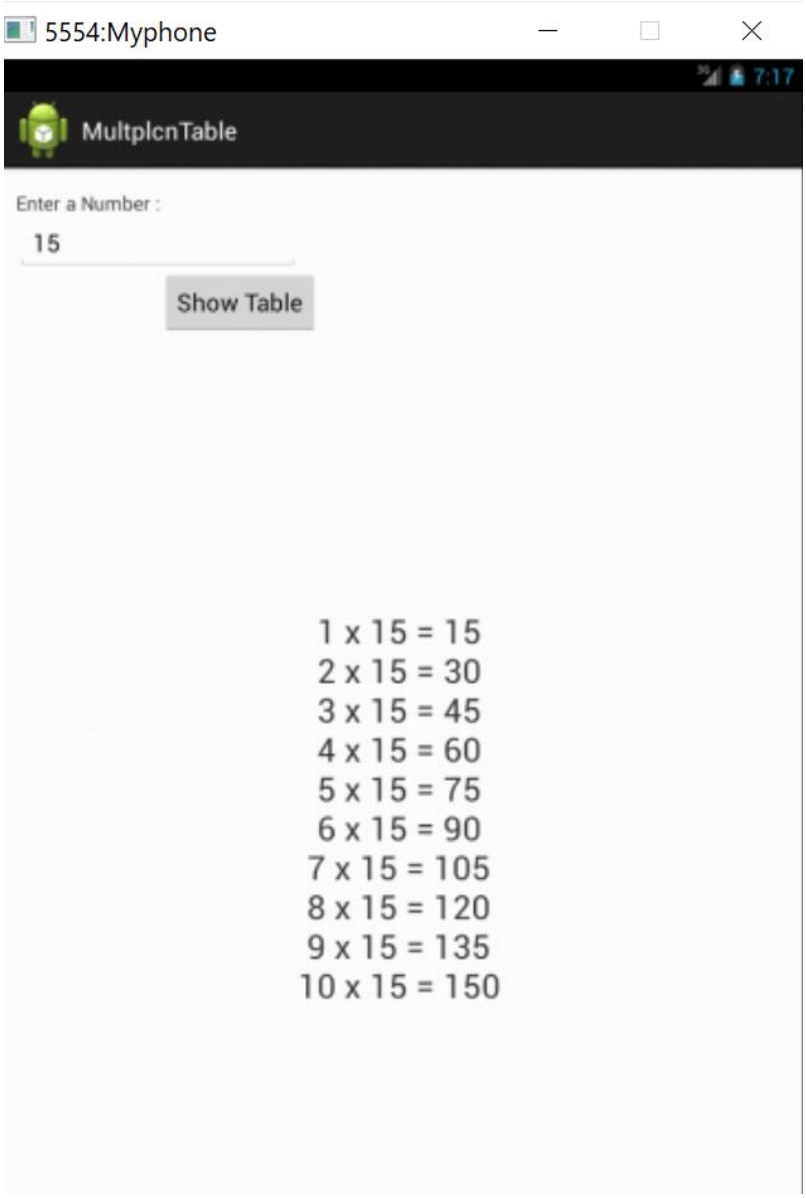
Xml File

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >
<TextView
    android:id="@+id/textView1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Enter a Number :" />
<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignLeft="@+id/textView1"
    android:layout_alignParentBottom="true"
    android:layout_alignParentRight="true"
    android:layout_below="@+id/button1"
    android:layout_marginTop="42dp"
    android:gravity="center"
    android:textSize="25sp" />
<EditText
    android:id="@+id/editText1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignLeft="@+id/textView1"
    android:layout_below="@+id/textView1"
    android:ems="10"
    android:inputType="number" />
<Button
    android:id="@+id/button1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@+id/editText1"
    android:layout_toRightOf="@+id/textView1"
    android:text="Show Table" />
</RelativeLayout>
```

Java File

```
package com.example.multplcntable;

import android.os.Bundle;
import android.app.Activity;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

    public class MainActivity extends Activity {
     EditText e1;
     Button b1;
     TextView t1;
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    e1=(EditText) findViewById(R.id.editText1);
    t1=(TextView) findViewById(R.id.textView2);
    b1=(Button) findViewById(R.id.button1);
    b1.setOnClickListener(new OnClickListener() {
            @Override
            public void onClick(View arg0) {
                // TODO Auto-generated method stub
                StringBuffer temp=new StringBuffer();
                String num=e1.getText().toString();
                int n=Integer.parseInt(num);
                for(int i=1;i<=10;i++)
                {
                    int ans=i*n;
                    temp.append(i +" x " + n + " = " + ans + "\n");
                    t1.setText(temp);
                }
            }
        });
}}
```



5. Check box (Write a Program to check the items listed)

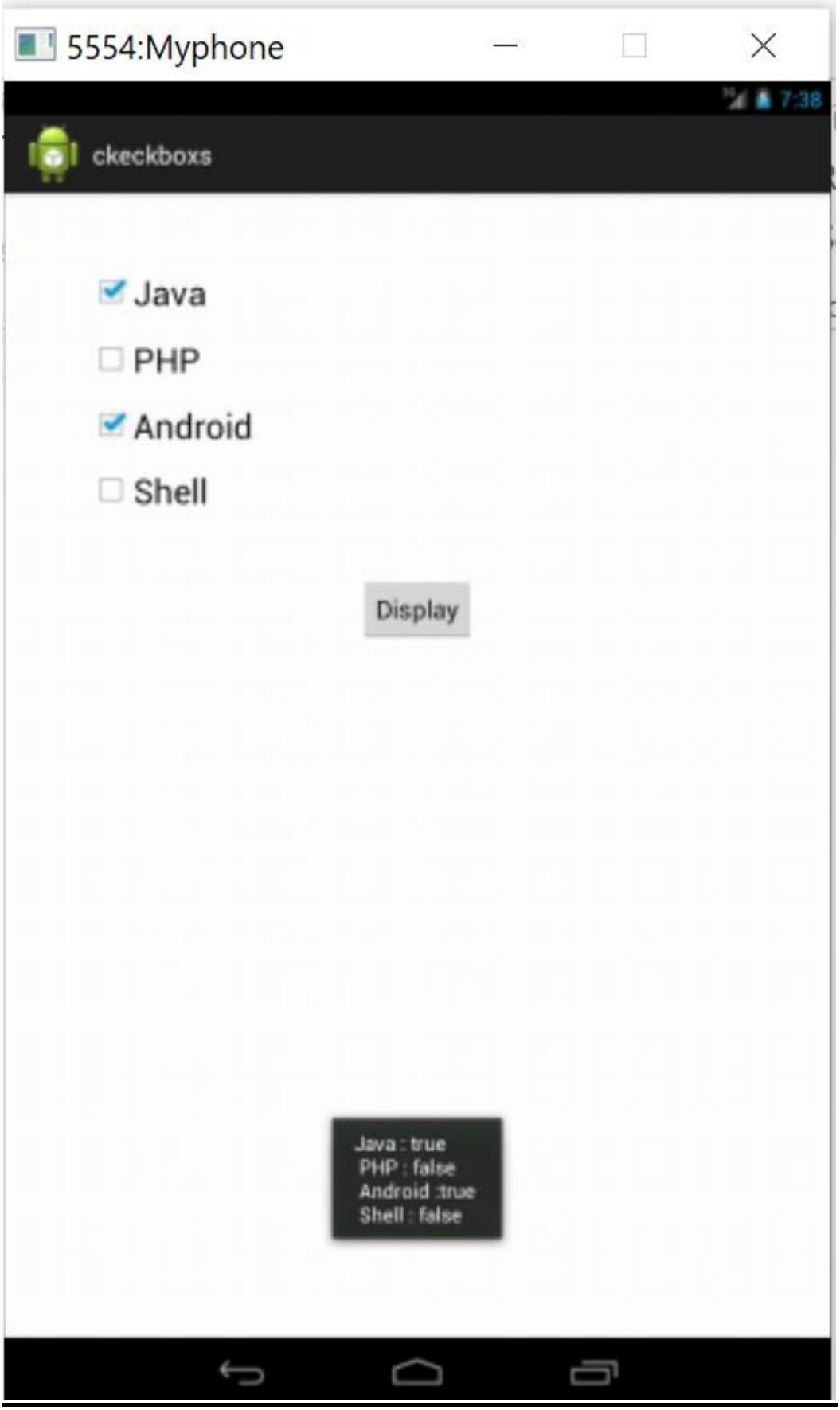
Xml File

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@+id/checkBox4"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="46dp"
        android:text="@string/bt" />

    <CheckBox
        android:id="@+id/checkBox1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentTop="true"
        android:layout_marginLeft="46dp"
        android:layout_marginTop="40dp"
        android:text="@string/jv"
        android:textSize="25sp" />

    <CheckBox
        android:id="@+id/checkBox2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignLeft="@+id/checkBox1"
        android:layout_below="@+id/checkBox1"
        android:layout_marginTop="16dp"
        android:text="@string/ph"
        android:textSize="25sp" />

    <CheckBox
        android:id="@+id/checkBox4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignLeft="@+id/checkBox3"
        android:layout_below="@+id/checkBox3"
        android:layout_marginTop="14dp"
        android:text="@string/os"
        android:textSize="25sp" />
    <CheckBox
        android:id="@+id/checkBox3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignLeft="@+id/checkBox2"
        android:layout_below="@+id/checkBox2"
        android:layout_marginTop="15dp"
        android:text="@string/Ad"
        android:textSize="25sp" />

</RelativeLayout>
```

Java File

```
package com.example.ckeckboxs;

import android.os.Bundle;
import android.app.Activity;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.Toast;

public class MainActivity extends Activity {
    Button bt;
    CheckBox cb1,cb2,cb3,cb4;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

     bt=(Button) findViewById(R.id.button1)    ;
     cb1=(CheckBox) findViewById(R.id.checkBox1);
     cb2=(CheckBox) findViewById(R.id.checkBox2);
     cb3=(CheckBox) findViewById(R.id.checkBox3);
     cb4=(CheckBox) findViewById(R.id.checkBox4);

     bt.setOnClickListener(new OnClickListener() {

            @Override
            public void onClick(View arg0) {
                  // TODO Auto-generated method stub

                  StringBuffer sb=new StringBuffer();
                  sb.append("Java : "  ).append(cb1.isChecked());
                  sb.append("\n PHP : ").append(cb2.isChecked());
                  sb.append("\n Android :").append(cb3.isChecked());
                  sb.append("\n Shell : ").append(cb4.isChecked());

                  Toast.makeText(MainActivity.this, sb.toString(),
Toast.LENGTH_LONG).show();



            }
       }) ;

    }

}
```

OUTPUT


5554:Myphone
7:38
ckeckboxs
Java
PHP
Android
Shell
Display
Java : true
PHP : false
Android :true
Shell : false

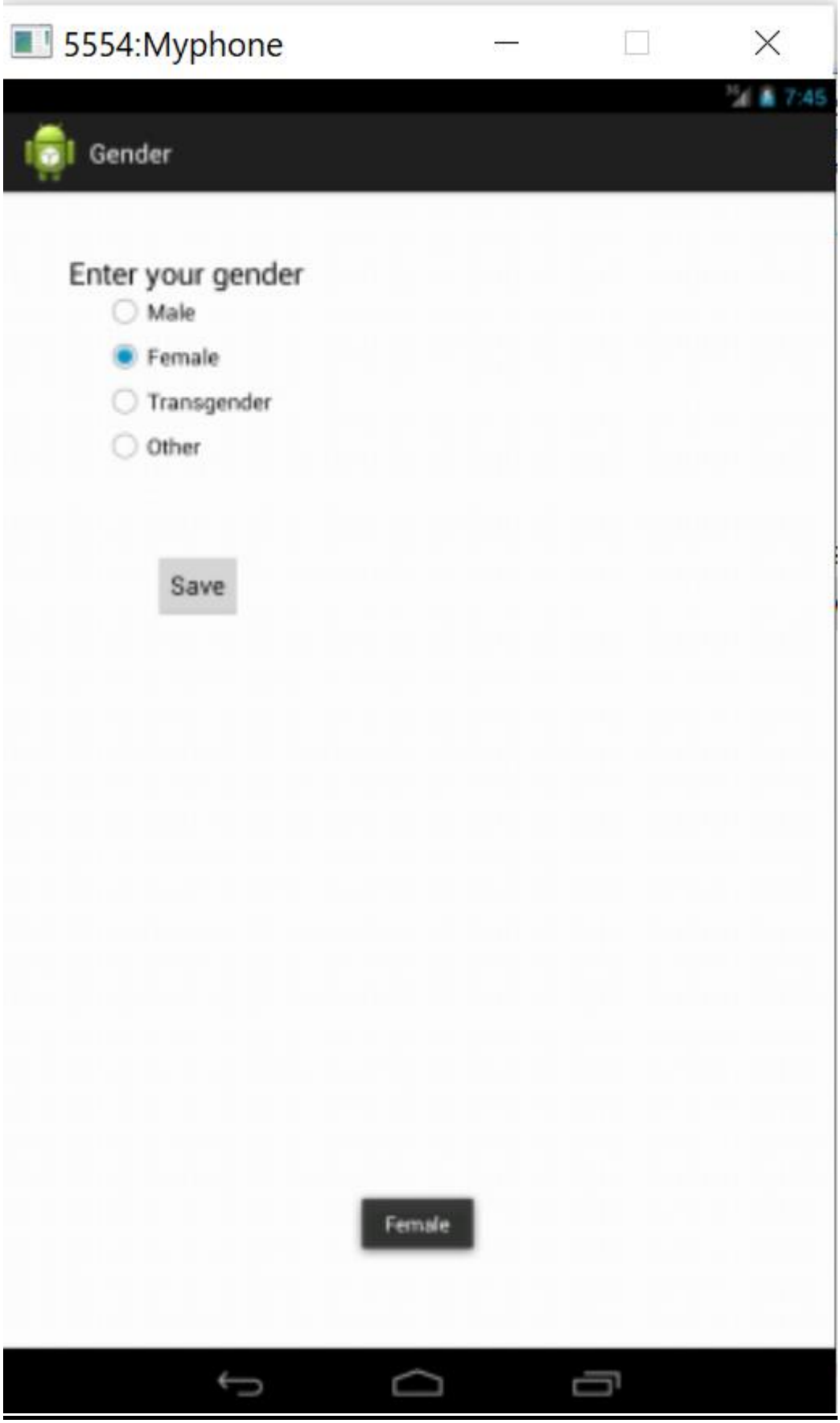
6 Radio Button (Write a Program to Select gender using radio button)

Xml File

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >
    <RadioGroup
        android:id="@+id/radioGroup1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentTop="true"
        android:layout_marginLeft="58dp"
        android:layout_marginTop="54dp" >
        <RadioButton
            android:id="@+id/radio0"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/m" />
        <RadioButton
            android:id="@+id/radio1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/f" />
        <RadioButton
            android:id="@+id/radio2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/t" />
        <RadioButton
            android:id="@+id/radio3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/o" />
    </RadioGroup>
    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignRight="@+id/radioGroup1"
        android:layout_below="@+id/radioGroup1"
        android:layout_marginRight="21dp"
        android:layout_marginTop="60dp"
        android:text="@string/s" />
    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentTop="true"
        android:layout_marginLeft="34dp"
        android:layout_marginTop="28dp"
        android:text="@string/st"
        android:textAppearance="?android:attr/textAppearanceLarge" />
</RelativeLayout>
```

Java File

```
package com.example.gender;

import android.os.Bundle;
import android.app.Activity;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.Toast;

public class MainActivity extends Activity {
Button btn;
RadioGroup rg;
RadioButton rb;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btn=(Button) findViewById(R.id.button1);
        rg=(RadioGroup) findViewById(R. id.radioGroup1);


        btn.setOnClickListener(new OnClickListener() {

        @Override
        public void onClick(View arg0) {
         // TODO Auto-generated method stub

         int selectd= rg.getCheckedRadioButtonId();
         rb= (RadioButton) findViewById(selectd);
         Toast.makeText(MainActivity.this, rb.getText(), 500).show();

          }
            });

    } }
```


5554:Myphone
7:45
Gender
Enter your gender
Male
Female
Transgender
Other
Save
Female

7. List View (Write a Program to Display the items in a list)

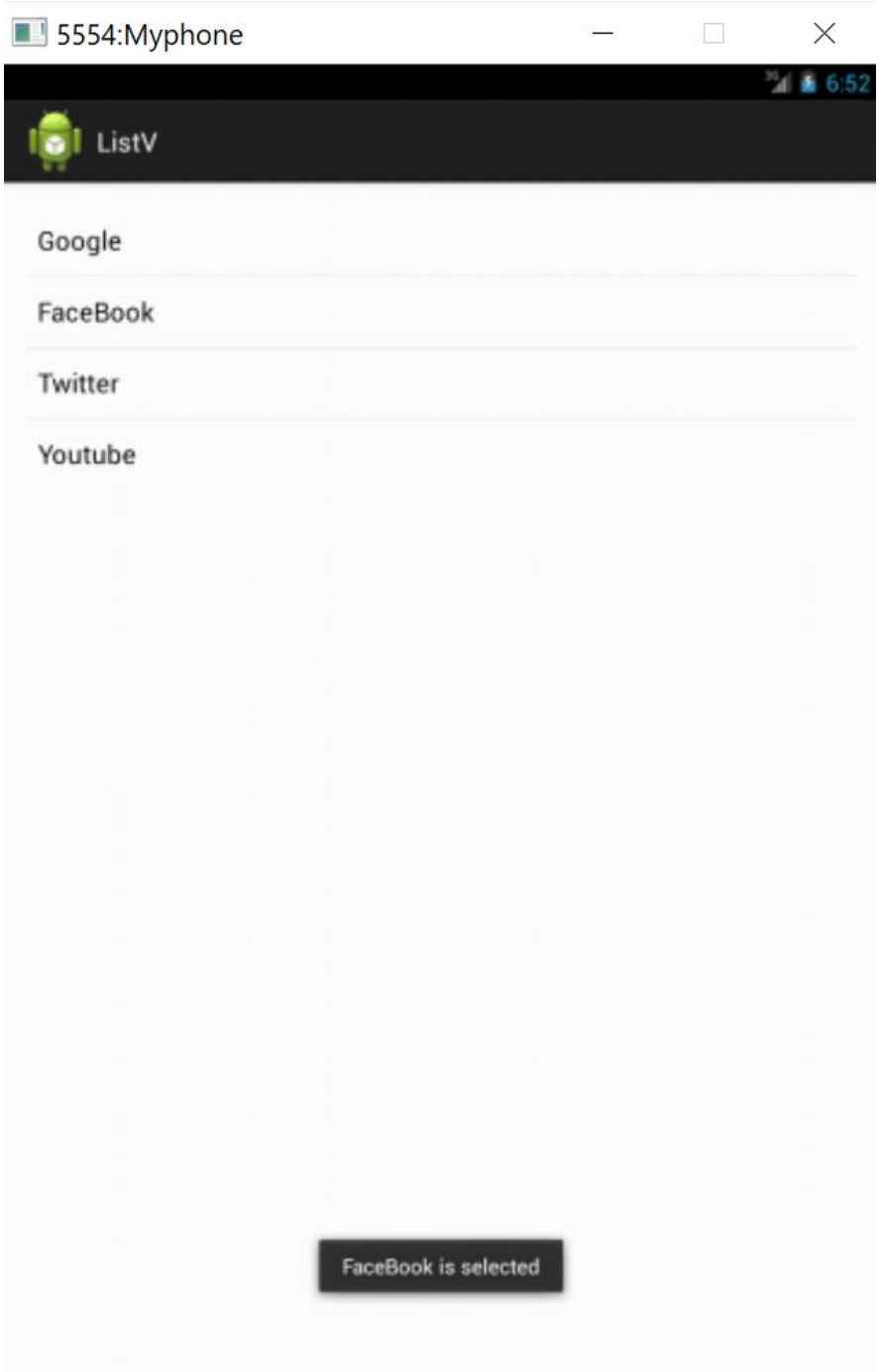
XML File

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >

    <ListView
        android:id="@+id/listView1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentTop="true" >
    </ListView>


    </RelativeLayout>
```

Java File

```
package com.example.listv;

import android.os.Bundle;
import android.app.Activity;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

public class MainActivity extends Activity implements OnItemClickListener  {
ListView lv;
String title[]={"Google","FaceBook","Twitter","Youtube"};
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        try
        {
           lv=(ListView) findViewById(R.id.listView1);
           ArrayAdapter<String> adt= new ArrayAdapter<String>(MainActivity.this,
android.R.layout.simple_list_item_1,title);
           lv.setAdapter(adt);
           lv.setOnItemClickListener(this);

        }
        catch(Exception e)
        {

            Toast.makeText(this,e.toString(),Toast.LENGTH_SHORT).show();
        }
        }
```

```
@Override
public void onItemClick(AdapterView<?>arg0,View arg1,int position,long
arg3)
      {
                    Toast.makeText(this, title[position]+" is selected",
                    Toast.LENGTH_LONG).show();
                     }
       }
```



8. Spinner (Write a Program to Spin the four items).

Xml File

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 android:paddingBottom="@dimen/activity_vertical_margin"
 android:paddingLeft="@dimen/activity_horizontal_margin"
 android:paddingRight="@dimen/activity_horizontal_margin"
 android:paddingTop="@dimen/activity_vertical_margin"
 tools:context=".MainActivity" >

 <Spinner
     android:id="@+id/spinner1"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:layout_alignParentTop="true"
     android:layout_centerHorizontal="true"
     android:layout_marginTop="24dp" />

 <TextView
     android:id="@+id/textView1"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:layout_alignParentBottom="true"
     android:layout_centerHorizontal="true"
     android:layout_marginBottom="138dp"
     android:text=" "
     android:textAppearance="?android:attr/textAppearanceLarge" />

 </RelativeLayout>
```

Java File

```
package com.example.spinner;

import android.os.Bundle;
import android.app.Activity;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemSelectedListener;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.TextView;

public class MainActivity extends Activity implements OnItemSelectedListener {
    Spinner sp;
    TextView tv;
    String [] items= {"Computer_Network", "Android_Programming",
"System_Software", "Operating-System"};
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        sp=(Spinner) findViewById(R.id.spinner1);
        tv= (TextView) findViewById(R.id.textView1);
```

```
    ArrayAdapter<String> ad= new ArrayAdapter<String>
    (this,android.R.layout.simple_list_item_1, items);
      sp.setAdapter(ad);
      sp.setOnItemSelectedListener(this);

  }
    @Override
    public void onItemSelected(AdapterView<?> arg0, View arg1, int arg2,
                long arg3) {
          // TODO Auto-generated method stub

          tv.setText(items[arg2]);

    }
    @Override
    public void onNothingSelected(AdapterView<?> arg0) {
          // TODO Auto-generated method stub

    }

}
```

9.Image View (Write a Program to Display images from local drive of the computer)

Xml File

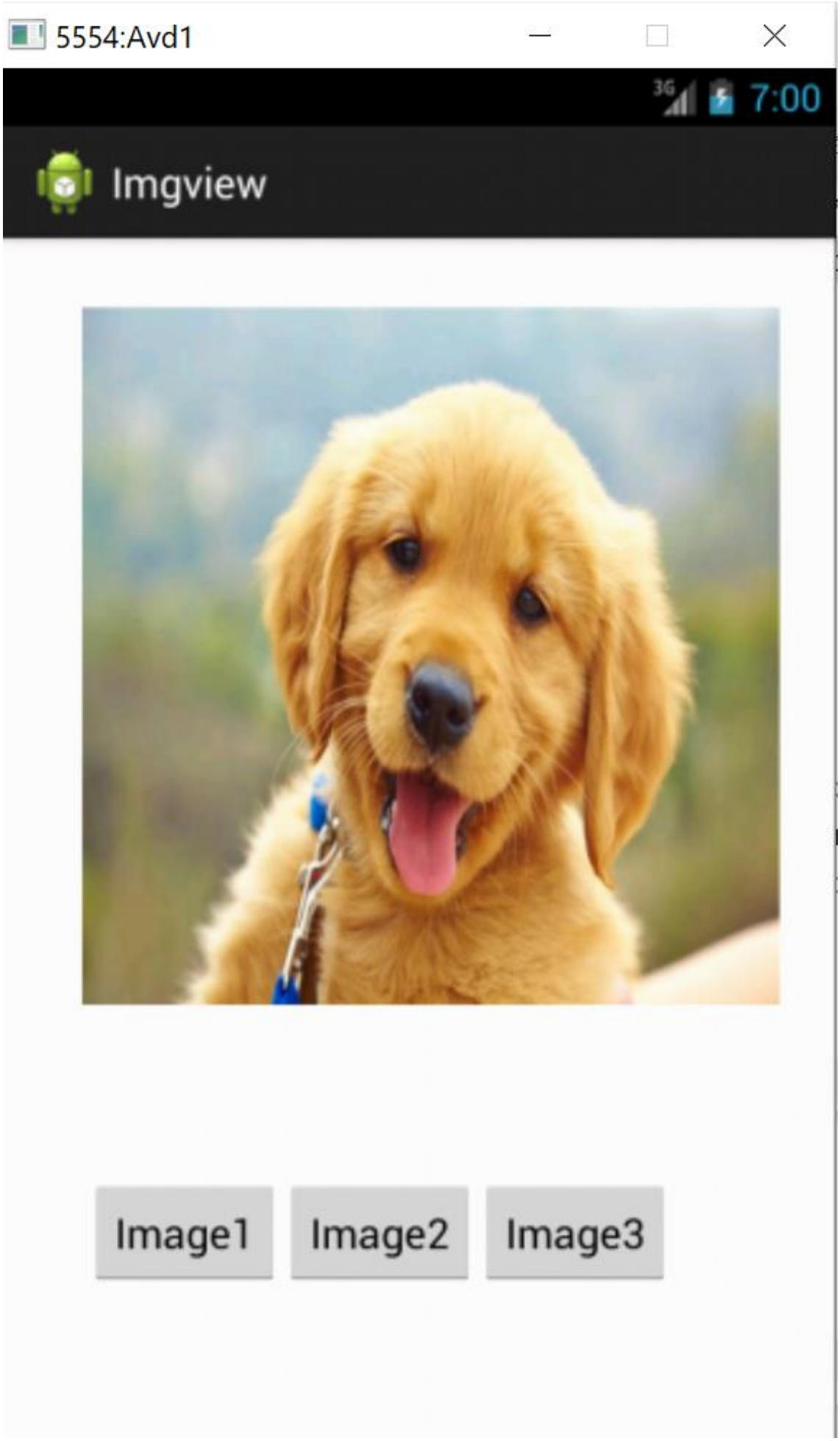
```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >

    <ImageView
        android:id="@+id/imageView1"
        android:layout_width="300dp"
        android:layout_height="300dp"
        android:layout_marginTop="14dp"
        android:layout_marginLeft="20dp"
         />

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentBottom="true"
        android:layout_alignParentLeft="true"
        android:layout_marginBottom="51dp"
        android:layout_marginLeft="22dp"
        android:text="@string/img1" />

    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignBaseline="@+id/button1"
        android:layout_alignBottom="@+id/button1"
        android:layout_toRightOf="@+id/button1"
        android:text="@string/img2" />

    <Button
        android:id="@+id/button3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignBottom="@+id/button2"
        android:layout_toRightOf="@+id/button2"
        android:text="@string/img3" />

</RelativeLayout>
```

Java File

```
package com.example.imgview;

import android.os.Bundle;
import android.app.Activity;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.ImageView;
```

```java
public class MainActivity extends Activity implements OnClickListener {

       Button b1,b2,b3;
      ImageView img;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

            b1=(Button) findViewById(R.id.button1);
            b2=(Button) findViewById(R.id.button2) ;
            b3=(Button) findViewById(R.id.button3);
            img=(ImageView) findViewById(R.id.imageView1);

            b1.setOnClickListener(this);
            b2.setOnClickListener(this);
            b3.setOnClickListener(this);

    }

      @Override
      public void onClick(View arg0) {
            // TODO Auto-generated method stub
            switch (arg0.getId())
            {
            case R.id.button1:
                  img.setBackgroundResource(R.drawable.image1);
                  break;
            case R.id.button2:
                  img.setBackgroundResource(R.drawable.image2);
                  break;
            case R.id.button3:
                  img.setBackgroundResource(R.drawable.image3);
                  break;
            default:
                  break;
            }
      }

}
```

5554:Myphone
7:07
Imgview
Image1
Image2
Image3


5554:Avd1
7:00
Imgview
Image1
Image2
Image3

10.Menu Program (Write a Program to create menu with three menu items)

Main.Xml File (Menu)

```
<menu xmlns:android="http://schemas.android.com/apk/res/android" >

    <item
        android:id="@+id/action_settings"
        android:orderInCategory="100"
        android:showAsAction="never"
        android:title="@string/action_settings"/>

    <item
        android:id="@+id/action_item1"
        android:orderInCategory="100"
        android:showAsAction="never"
        android:title="@string/action_item1"/>
    <item
        android:id="@+id/action_item2"
        android:orderInCategory="100"
        android:showAsAction="never"
        android:title="@string/action_item2"/>
    <item
        android:id="@+id/action_item3"
        android:orderInCategory="100"
        android:showAsAction="never"
        android:title="@string/action_item3"/>

</menu>
```

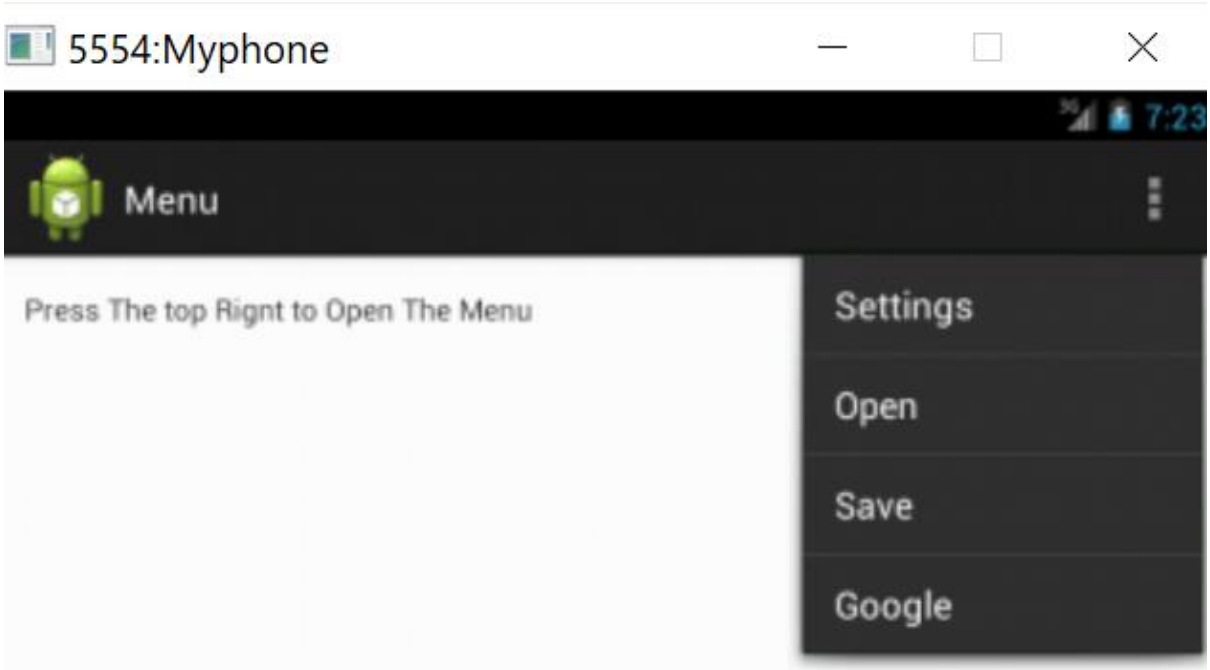
Layout.Xml File

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >


    <TextView android:layout_height="wrap_content"
        android:layout_width="wrap_content"
        android:id="@+id/textView1"
         android:text="Press The top Rignt to Open The Menu"/>

</RelativeLayout>
```

Java File

```
 package com.example.menu;

import android.net.Uri;
import android.os.Bundle;
import android.app.Activity;
import android.content.Intent;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
```

```java
public class MainActivity extends Activity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }


    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
                  getMenuInflater().inflate(R.menu.main, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
      // TODO Auto-generated method stub
      int id= item.getItemId();
      switch (id)
      {
      case R.id.action_item1:
            Toast.makeText(getApplicationContext(), "File Opened",
Toast.LENGTH_LONG).show();
      break;
      case R.id.action_item2:
            Toast.makeText(getApplicationContext(), "File Saved",
Toast.LENGTH_LONG).show();
      break;
      case R.id.action_item3:
            //Toast.makeText(getApplicationContext(), "File can be Edited",
Toast.LENGTH_LONG).show();
            Intent i=new Intent(Intent.ACTION_VIEW);
            i.setData(Uri.parse("http://www.google.com"));
            startActivity(i);
      break;
      }
      return false;

      //return super.onOptionsItemSelected(item);
    }
}
```


5554:Myphone
7:23
Menu
Press The top Rignt to Open The Menu
Settings
Open
Save
Google

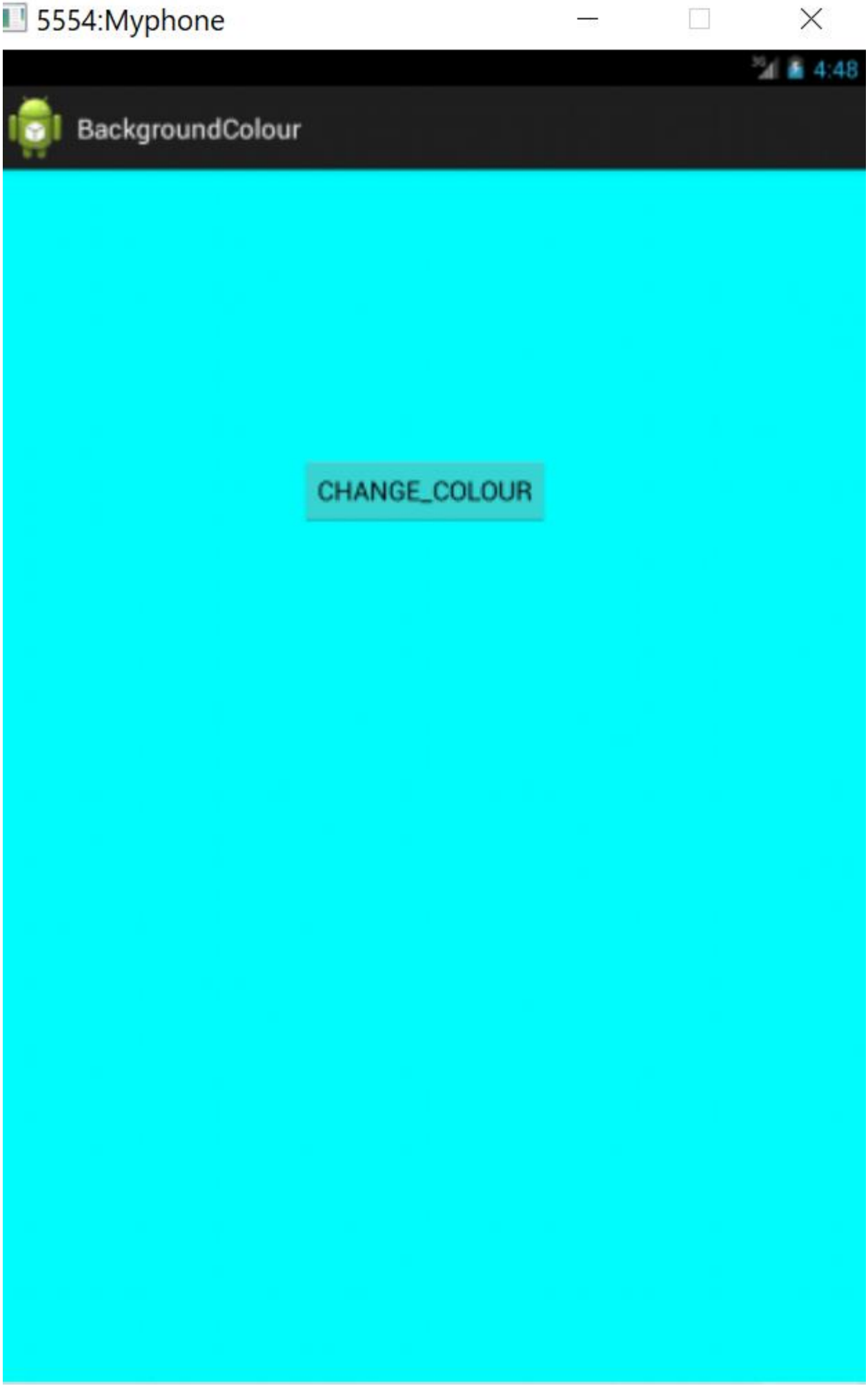
11. Write a program to Change the Background colour of the Activity

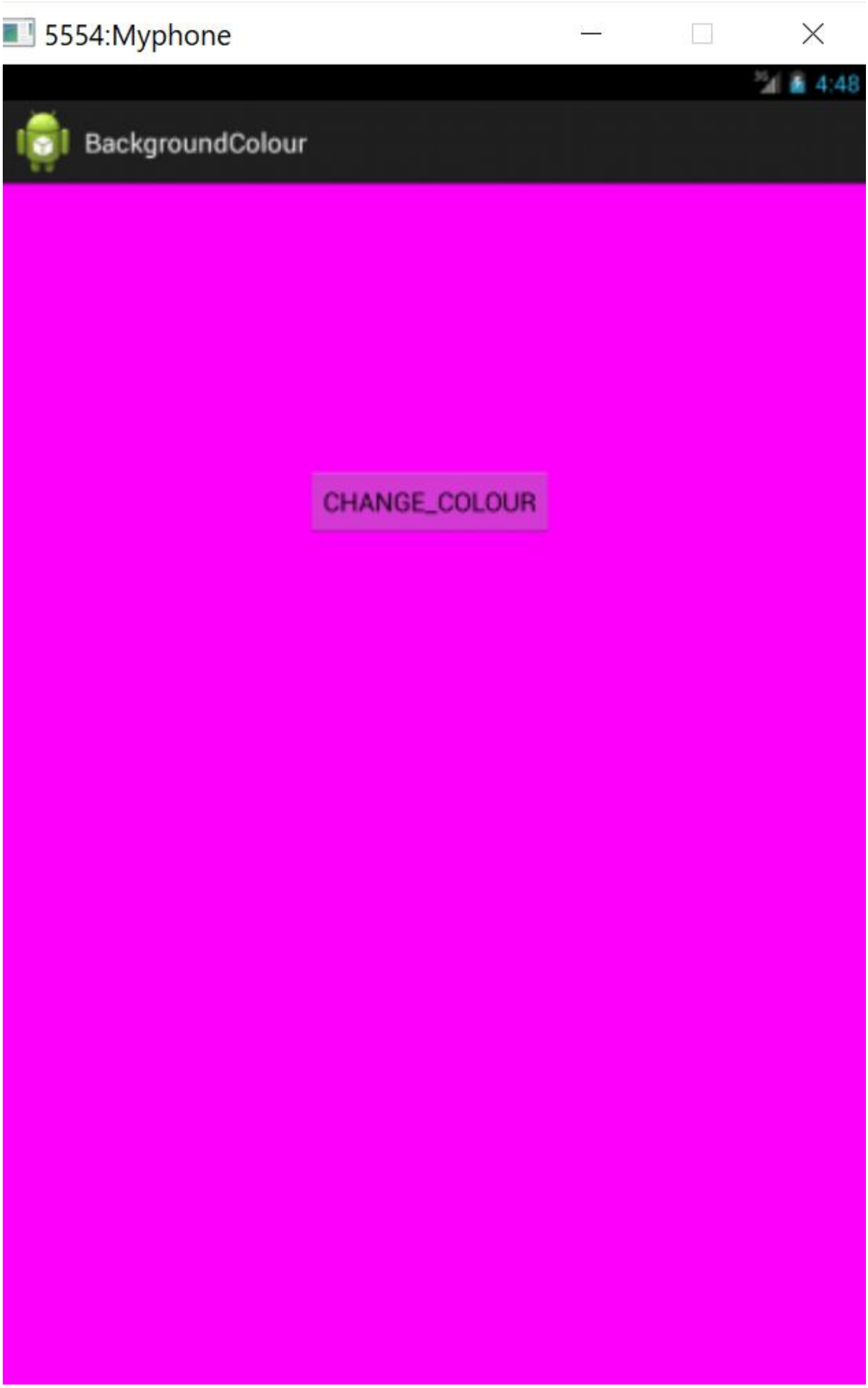
Xml File

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/r1"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >

<Button
    android:id="@+id/button2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignBottom="@+id/button1"
    android:layout_toRightOf="@+id/button1"
    android:text="CHANGE COLOR" />

</RelativeLayout>
```

Java File

```
package com.gems.pqr;

import android.os.Bundle;
import android.app.Activity;
import android.graphics.Color;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.RelativeLayout;
public class MainActivity extends Activity {
      Button b1;
      RelativeLayout rel;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        b1=(Button) findViewById(R.id.button1);

        rel=(RelativeLayout) findViewById(R.id.r1);
        rel.setBackgroundColor(Color.CYAN);
        b1.setOnClickListener(new OnClickListener() {

                @Override
                public void onClick(View arg0) {
                      // TODO Auto-generated method stub
                      rel.setBackgroundColor(Color.MAGENTA);
                }
            });

    }
}
```


5554:Myphone
4:48
BackgroundColour
CHANGE_COLOUR


5554:Myphone
4:48
BackgroundColour
CHANGE_COLOUR

12. Write a program to start another activity from your own activity using intent

First Xml File

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#0000ff"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >
<TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentTop="true"
        android:layout_marginTop="30dp"
        android:text="Now you are in firstpage "
        android:textAppearance="?android:attr/textAppearanceLarge" />

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@+id/textView1"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="36dp"
        android:text="Go to next page" />
</RelativeLayout>
```

Second Xml File

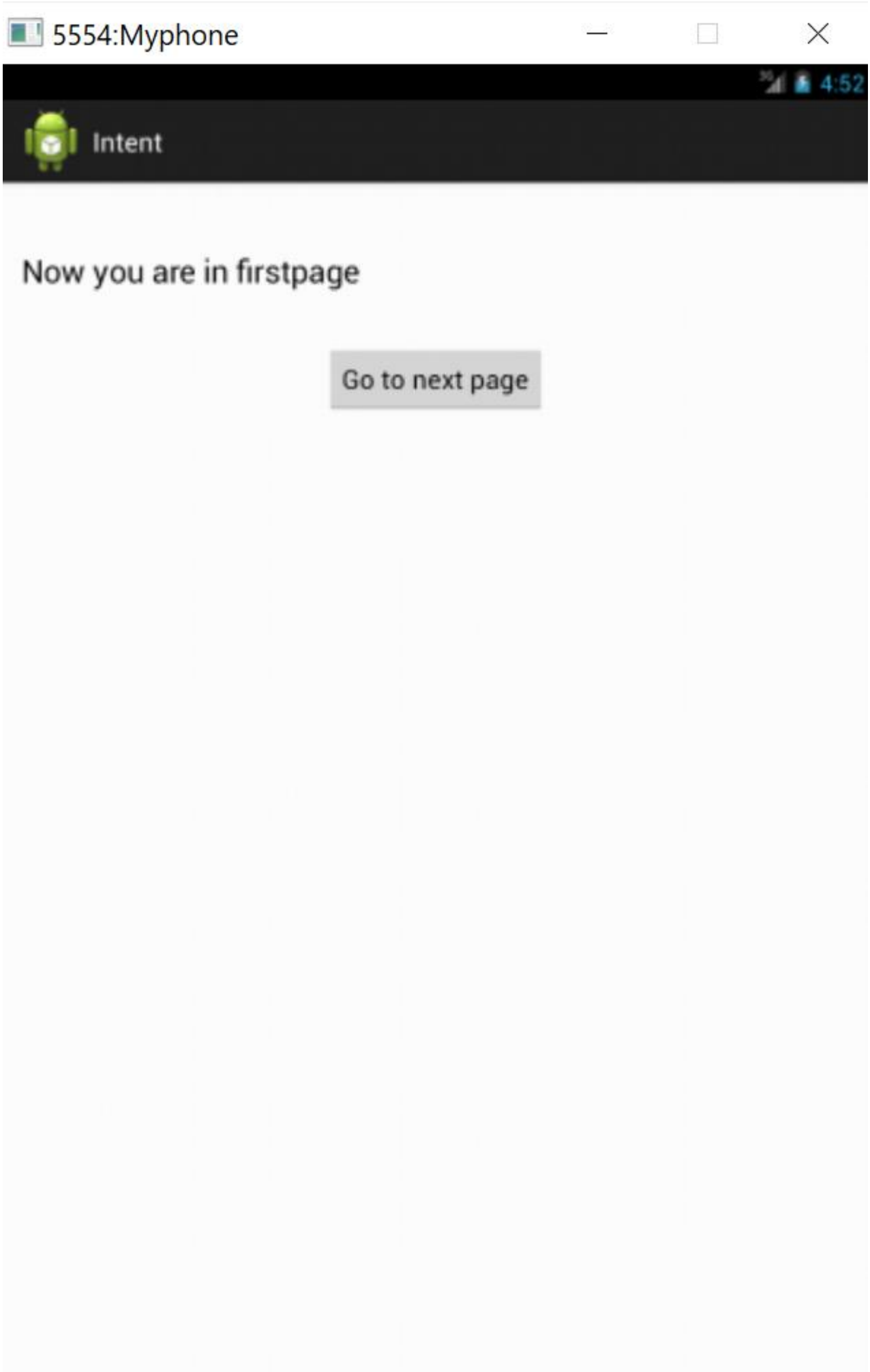
```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#ff0000"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".SecondActivity" >

<TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentTop="true"
        android:layout_marginLeft="46dp"
        android:layout_marginTop="59dp"
        android:text="Now you are in secondpage"
        android:textAppearance="?android:attr/textAppearanceLarge" />
    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignLeft="@+id/textView1"
        android:layout_below="@+id/textView1"
        android:layout_marginTop="52dp"
        android:text="Go to previous Activity" />
</RelativeLayout>
```

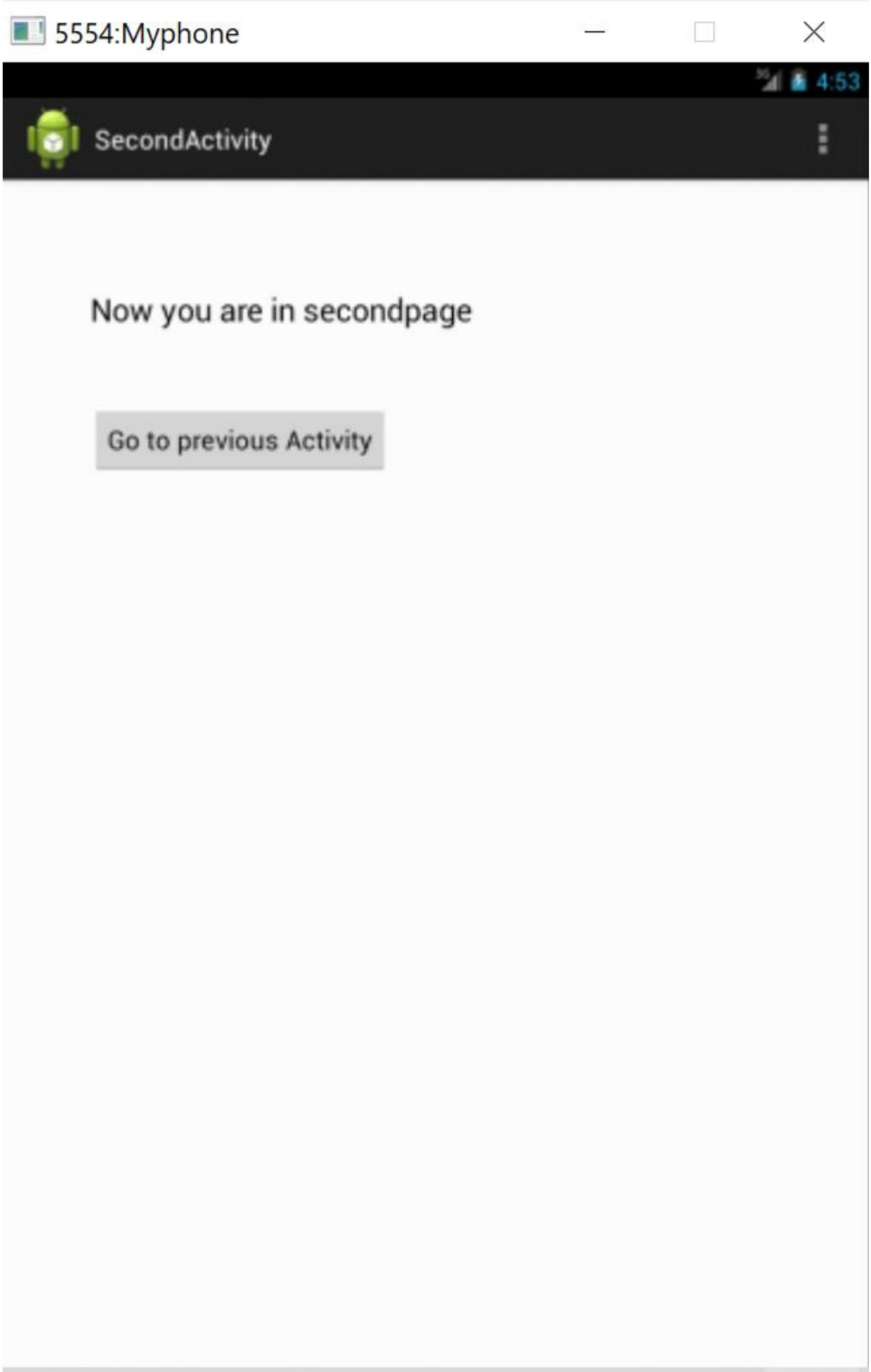
First Java File

```
package com.andoubleos.usingintent;

import android.os.Bundle;
import android.app.Activity;
import android.content.Intent;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;

public class MainActivity extends Activity implements OnClickListener {
      Button btn;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btn=(Button) findViewById(R.id.button1);
        btn.setOnClickListener(this);
    }

      @Override
      public void onClick(View arg0) {
            // TODO Auto-generated method stub
            Intent i = new Intent(MainActivity.this,SecondActivity.class);
            startActivity(i);
      }
}
```

Second Java File

```
package com.andoubleos.usingintent;

import android.os.Bundle;
import android.app.Activity;
import android.content.Intent;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;

public class SecondActivity extends Activity implements OnClickListener {
      Button btn;

      @Override
      protected void onCreate(Bundle savedInstanceState) {
            super.onCreate(savedInstanceState);
            setContentView(R.layout.activity_second);

             btn=(Button) findViewById(R.id.button1);
           btn.setOnClickListener(this);
      }

      @Override
      public void onClick(View arg0) {
            // TODO Auto-generated method stub
            Intent i = new Intent(SecondActivity.this,MainActivity.class);
            startActivity(i);
      }
}
```


5554:Myphone
4:52
Intent
Now you are in firstpage
Go to next page


5554:Myphone
4:53
SecondActivity
Now you are in secondpage
Go to previous Activity

13. Alert Box (Write a program to Display an alert box with OK and Cancel)

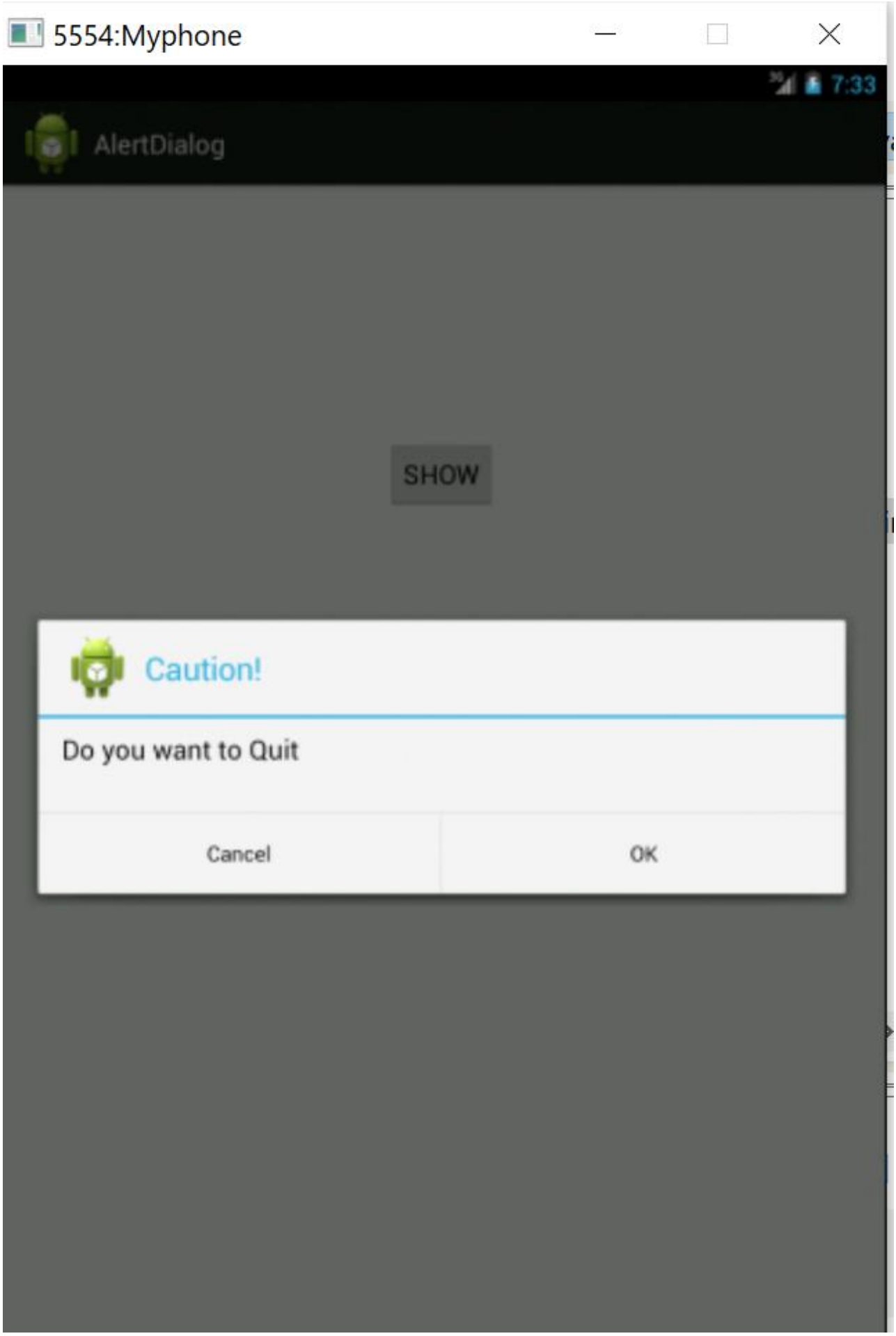
Xml File

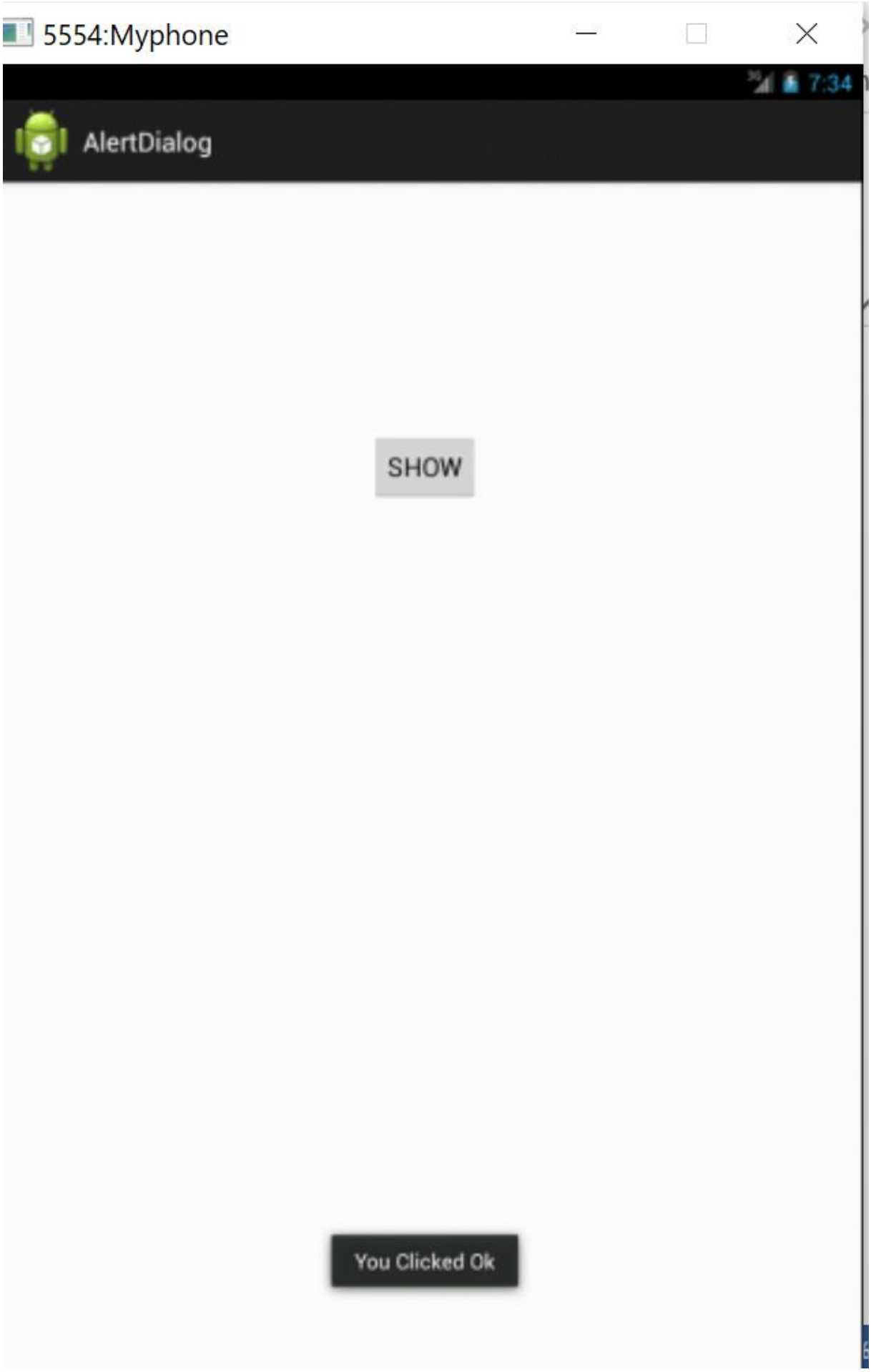
```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 android:paddingBottom="@dimen/activity_vertical_margin"
 android:paddingLeft="@dimen/activity_horizontal_margin"
 android:paddingRight="@dimen/activity_horizontal_margin"
 android:paddingTop="@dimen/activity_vertical_margin"
 tools:context=".MainActivity" >



 <Button
     android:id="@+id/button1"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:layout_alignParentTop="true"
     android:layout_centerHorizontal="true"
     android:layout_marginTop="156dp"
     android:text="@string/show" />

</RelativeLayout>
```

Java File

```
  package com.example.alertdialog;

import android.os.Bundle;
import android.app.Activity;
import android.app.AlertDialog;
import android.content.DialogInterface;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends Activity implements OnClickListener {
      Button bt1;

    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        bt1=(Button) findViewById(R.id.button1);
        bt1.setOnClickListener(this);
    }

   public void onClick(View arg0)
    {
            AlertDialog.Builder b1= new AlertDialog.Builder(MainActivity.this);
                        b1.setIcon(R.drawable.ic_launcher);
                        b1.setTitle("Caution!");
                        b1.setMessage("Do you want to Quit");
```

```
b1.setPositiveButton("OK", new DialogInterface.OnClickListener()
      {

         @Override
         public void onClick(DialogInterface arg0, int arg1)
           {
Toast.makeText(MainActivity.this, "You Clicked Ok",Toast.LENGTH_LONG).show();

            }
       });
 b1.setNegativeButton("Cancel", new DialogInterface.OnClickListener()
       {

        @Override
        public void onClick(DialogInterface arg0, int arg1)
            {
Toast.makeText(MainActivity.this,"You Clicked Cancel", Toast.LENGTH_LONG).show();

            }
        });
 AlertDialog d1= b1.create();
 d1.show();

             }
        }
```


5554:Myphone
7:33
AlertDialog
SHOW
Caution!
Do you want to Quit
Cancel
OK


5554:Myphone
7:34
AlertDialog
SHOW
You Clicked Ok

14. Write a program to Change the Background image

Xml File

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/rView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >
    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentBottom="true"
        android:layout_centerHorizontal="true"
        android:layout_marginBottom="47dp"
        android:text="CHANGE" />

</RelativeLayout>
```

Java File

```
package com.andoubleos.background;

import java.util.Random;

import android.os.Bundle;
import android.app.Activity;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;

public class MainActivity extends Activity {

      Button btn;
      View screenView;
      int[] images;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btn =(Button) findViewById(R.id.button1);
        screenView = findViewById(R.id.rView);

        images = new int[] { R.drawable.image1,R.drawable.image2,
                        R.drawable.image3, R.drawable.image4,R.drawa ble.image6};
        btn.setOnClickListener(new OnClickListener() {

                  @Override
                  public void onClick(View arg0) {
                        // TODO Auto-generated method stub
                        int aryLength = images.length;
                        Random rand = new Random();
                        int randNum = rand.nextInt(aryLength);
                        screenView.setBackgroundResource(images[randNum]);

                  }
            });
    }}
```

5554:Myphone
4:45
BGImage_Random
CHANGE

5554:Myphone
4:46
BGImage_Random
CHANGE

15. Write a program to Display Date and Time using Dialog Box

Xml File

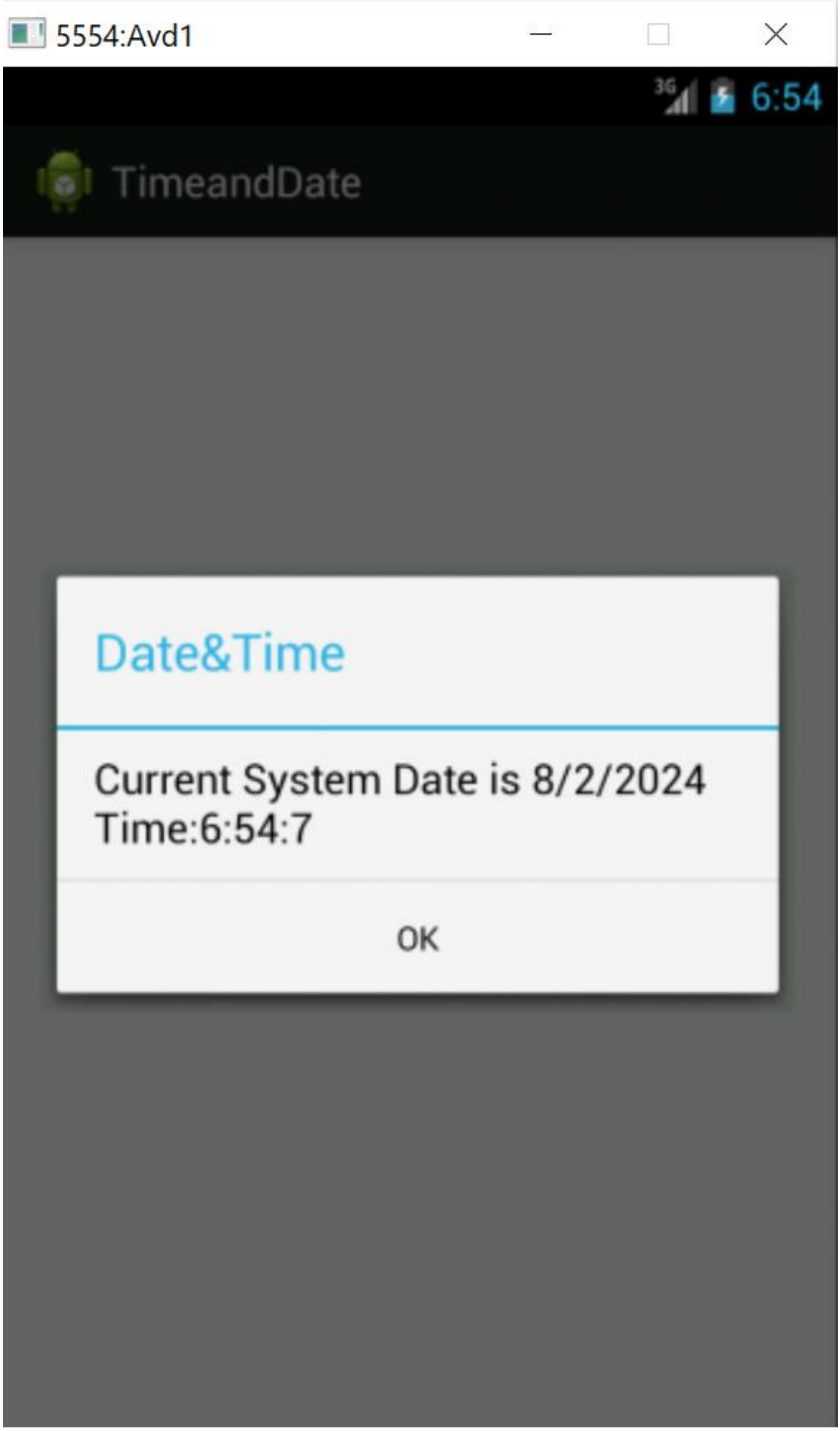
```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentTop="true"
        android:layout_marginLeft="104dp"
        android:layout_marginTop="154dp"
        android:text="Display Date" />

</RelativeLayout>
```

Java File

```
package com.example.timeanddate;

import java.util.Calendar;
import android.os.Bundle;
import android.app.Activity;
import android.app.AlertDialog;
import android.content.DialogInterface;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import java.util.Calendar;
public class MainActivity extends Activity {
Button bt;
 int year, month, day,hr,min,sec;
 String sysdate,systime;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        bt= (Button) findViewById(R.id.button1);
        bt.setOnClickListener(new OnClickListener() {

                    @Override
                    public void onClick(View arg0) {
                          // TODO Auto-generated method stub
                          Calendar cal= Calendar.getInstance();
                          year = cal.get(Calendar.YEAR);
                          month = cal.get(Calendar.MONTH);
                          day = cal.get(Calendar.DAY_OF_MONTH);
                          hr =cal.get(Calendar.HOUR);
                          min=cal.get(Calendar.MINUTE);
                          sec=cal.get(Calendar.SECOND);
```

```
                        sysdate = day +"/"+(month+1)+"/"+year;
                        systime= hr+":"+min+":"+sec;

AlertDialog.Builder  b1= new AlertDialog.Builder(MainActivity.this);
b1.setTitle("Date&Time");
b1.setMessage("Current System Date is "+sysdate+"\nTime:"+systime );
b1.setPositiveButton("OK", new DialogInterface.OnClickListener() {

@Override
public void onClick(DialogInterface arg0, int arg1) {
// TODO Auto-generated method stub



  }
});
AlertDialog d= b1.create();
d.show();

      }
            });
    }


 }
```

5554:Avd1
6:53
TimeandDate
Display Date


5554:Avd1
6:54
TimeandDate
Date&Time
Current System Date is 8/2/2024
Time:6:54:7
OK